

TOPIC EIGHT - AIR-CONDITIONING SYSTEMS

1. GENERAL INTRODUCTION.

Air conditioning, often abbreviated as A/C or AC, is the process of removing heat and controlling the humidity of air in an enclosed space to achieve a more comfortable interior environment by use of powered 'air conditioners' or a variety of other methods including passive cooling and ventilative cooling.

Most office buildings require an air-conditioning (AC) system to control temperature and provide a stable and comfortable working environment. In certain environments, AC systems are required to provide more than basic temperature control. They are also used to give control over the humidity and cleanliness of the air supply. For example, this applies to hospital operating theatres, pharmaceutical production areas and specific industrial clean room applications.

Air conditioning may be required for the following reasons.

- Thermal comfort of the occupants will increase productivity, improve the health of the workers and reduce absenteeism.
- The space conditions may need to be controlled for a particular process. In this case human comfort may not be met.
- Set conditions of air purity, moisture content, and temperature may need to be maintained. This is necessary in clean rooms, hospital operating theatres, museums, stores for old manuscripts, computer rooms and the like

Air conditioning of a space is necessary if space conditions have to be held at a constant value, regardless of the external conditions. An example would be a computer room, which has to be held at fixed temperature and moisture content regardless of the external conditions.

Consider a room that has to be held at 20 °C when the external air temperature is 28 °C. Without air cooling the space cannot be held at 20 °C, because the interior and exterior heat gains mean that the space temperature would always be higher than the external conditions.

Ventilation will remove some of these internal gains. The quantity of air necessary for cooling will vary, depending on many factors. If a large temperature difference exists between the supply air and the room air, only a small air quantity will be required; if a small temperature difference between the supply and room air exists, a larger air quantity becomes necessary. The air quantity has a direct effect on the size of the ductwork and hence the space taken up in horizontal or vertical runs. The heat gains to be considered are:

- **Sensible heat** - the gains from occupants, lights, machines, processes, infiltration, conduction and solar, through the windows and other parts of the fabric;
- **Latent heat** - the moisture gains from infiltration, occupants and processes.

The sensible gains have a direct influence on the air temperature, while the latent gains do not directly influence the surrounding temperature, but influence the moisture content in the space.

It is essential to ensure that a simple ventilation system is not classed as an air-conditioning system. An air-conditioning system must be capable of achieving the following:

- filtering (air purity);
- heating;
- cooling (without moisture removal);
- humidification (moisture addition);
- dehumidifying (moisture removal);
- distributing the air within the space without creating discomfort or stagnant pockets.

2. TERMINOLOGY USED IN AIR CONDITIONING DESIGN:

To effectively comprehend air conditioning design, it is important to familiarize yourself with the pertinent terminology. Some key terms include:

- i. **HVAC (Heating, Ventilation, and Air Conditioning):** refers to the overall system that provides both heating and cooling, along with ventilation, in a building. It encompasses various components, such as heating units, air conditioners, ventilation fans, and ductwork, working together to create a comfortable indoor environment.
- ii. **BTU (British Thermal Unit):** is a unit of measurement used to quantify the amount of heat energy required to raise or lower the temperature of a given amount of air. It is commonly used to determine the cooling or heating capacity of air conditioning systems.
- iii. **Refrigeration Cycle:** is the process through which air conditioning systems remove heat from indoor spaces and release it outside. It involves the circulation of a refrigerant, which changes state from a gas to a liquid and back to a gas, absorbing and releasing heat in the process.
- iv. **Sensible Heat:** refers to the heat energy that causes a change in temperature without a state change. In air conditioning, sensible heat refers to the heat added or removed from the air, resulting in a change in its temperature.

- v. **Heat Exchanger:** is a device used in air conditioning systems to transfer heat between two fluids without them coming into direct contact. It allows for the exchange of thermal energy between the indoor and outdoor environments.
- vi. **Compressor:** is a vital component of an air conditioning system that pressurizes the refrigerant gas, raising its temperature and pressure. It plays a crucial role in the refrigeration cycle by circulating the refrigerant through the system.
- vii. **Condenser:** a heat exchanger located in the outdoor unit of an air conditioning system. It receives the high-pressure, high-temperature refrigerant gas from the compressor and releases heat to the outside air, causing the refrigerant to condense into a liquid state.
- viii. **Evaporator:** is another heat exchanger, typically located in the indoor unit of an air conditioning system. It receives the low-pressure, low-temperature liquid refrigerant from the expansion valve and allows it to evaporate, absorbing heat from the indoor air in the process.
- ix. **Expansion Valve:** is a device that regulates the flow of refrigerant into the evaporator. It reduces the pressure of the refrigerant, allowing it to expand and evaporate, thus absorbing heat from the indoor air.
- x. **Ductwork:** refers to the network of channels or pipes that distribute conditioned air throughout a building. It plays a crucial role in delivering cool or warm air to different rooms and maintaining proper airflow.
- xi. **Thermostat:** is a control device used to regulate and maintain the desired temperature in a building. It allows users to set the temperature and control the operation of the air conditioning system.
- xii. **Psychometrics:** is the study of the properties of air and its moisture content. It involves understanding concepts such as dry-bulb temperature, wet-bulb temperature, relative humidity, and dew point, which are crucial in air conditioning design.

3. PSYCHROMETRICS AND AIR PROPERTIES:

Psychometrics is the study of the physical and thermodynamic properties of air and their relationship to moisture content. Understanding psychometrics is crucial in the design and operation of air conditioning systems as it allows us to analyze and manipulate air properties to achieve desired indoor conditions.

Here are some key parameters and concepts related to psychometrics:

a) **Dry-Bulb Temperature:**

The dry-bulb temperature represents the temperature of the air measured by a regular thermometer. It is an essential parameter in determining the sensible heat transfer in an air conditioning system. Sensible heat refers to the heat that causes a change in the air's temperature without changing its moisture content.

b) Wet-Bulb Temperature:

The wet-bulb temperature is the lowest temperature that can be achieved by evaporating water into the air at constant pressure. It is an indicator of the air's moisture content and is used to evaluate the cooling potential of the system. The wet-bulb temperature is measured using a thermometer with its bulb covered in a wetted wick or fabric.

c) Relative Humidity (RH):

Relative humidity is the ratio of the actual amount of moisture present in the air to the maximum amount it can hold at a given temperature. It is expressed as a percentage. Relative humidity indicates the air's moisture saturation level and plays a significant role in occupant comfort. High relative humidity can make the air feel damp and uncomfortable, while low relative humidity can lead to dryness and potential health issues.

d) Dew Point:

The dew point is the temperature at which the air becomes saturated with moisture, causing condensation to occur. It is an important parameter to consider for preventing condensation-related issues, such as the formation of moisture on surfaces or within the air conditioning system. Dew point is influenced by both the air's temperature and its moisture content.

4. SELECTION CRITERIA FOR AIR-CONDITIONING SYSTEM

Selecting an appropriate air conditioning system for a building is crucial to ensure optimal performance and energy efficiency. There are several factors to consider when making this decision. Let's explore some key selection criteria:

a) Cooling Load Calculation:

The cooling load calculation is a fundamental step in determining the capacity of the air conditioning system needed for a specific space. It takes into account factors such as the size of the area, heat gain from sunlight, occupancy, equipment, and insulation levels. Properly calculating the cooling load helps ensure that the system can effectively cool the space without being undersized or oversized.

Example: Let's say we have a 500-square-foot office space with large windows facing west, resulting in significant solar heat gain. The cooling load calculation would consider the area's size, the heat gain from the windows, the number of occupants, and other factors to

determine the appropriate capacity of the air conditioning system needed to maintain a comfortable temperature.

b) Energy Efficiency:

Energy efficiency is a crucial consideration when selecting an air conditioning system. Look for systems with high Seasonal Energy Efficiency Ratio (SEER) ratings for cooling or Heating Seasonal Performance Factor (HSPF) ratings for heating. Higher ratings indicate greater energy efficiency, resulting in lower operating costs and reduced environmental impact.

Example: An air conditioning system with a SEER rating of 18 consumes less energy to provide the same cooling output compared to a system with a SEER rating of 14. By choosing the more efficient system, you can save on energy costs and contribute to a greener environment.

c) Noise Levels:

Consider the noise levels produced by the air conditioning system, especially if it will be installed in areas where low noise is essential, such as bedrooms or offices. Look for systems with lower decibel ratings, ensuring a quieter and more comfortable environment.

Example: In a bedroom, a quieter air conditioning system with a decibel rating of 50 will create a more peaceful and conducive sleeping environment compared to a system with a rating of 70, which may cause disturbances.

d) Maintenance Requirements:

Take into account the maintenance requirements of the air conditioning system. Some systems may require regular filter changes, coil cleaning, or professional servicing. Consider the ease of maintenance and availability of service technicians when selecting a system.

Example: A system with easily accessible filters and clear maintenance instructions may be more convenient to maintain compared to a system that requires complex disassembly for routine maintenance tasks.

e) Environmental Considerations:

Consider the environmental impact of the air conditioning system. Look for systems that use refrigerants with lower Global Warming Potential (GWP) and ozone depletion potential. Additionally, consider systems that incorporate environmentally friendly features such as energy-saving modes and programmable thermostats.

Example: Choosing an air conditioning system that uses refrigerants with lower GWP, such as R-410A, instead of older refrigerants like R-22, helps reduce the system's impact on global warming and ozone depletion.

f) Air Distribution:

Evaluate the air distribution capabilities of the system. Consider factors such as the layout of the space, the number and placement of diffusers or vents, and the ability of the system to deliver conditioned air evenly throughout the room. Proper air distribution ensures consistent comfort levels in different areas of the building.

Example: In a large open-plan office space, a system with strategically placed diffusers and adjustable airflow control can provide effective air distribution, preventing hot or cold spots and ensuring a comfortable working environment for all occupants.

g) Zoning Options:

Consider whether the air conditioning system offers zoning capabilities. Zoning allows for independent temperature control in different areas or zones of a building. This feature is particularly useful in multi-story buildings or spaces with varying occupancy levels, as it allows customized comfort and energy efficiency.

Example: In a residential building with multiple floors, zoning capabilities enable different temperature settings for each floor, allowing occupants to personalize their comfort while optimizing energy usage.

h) Air Quality Enhancement:

Assess the air quality enhancement features of the system. Look for systems that incorporate filters or air purifiers to remove dust, allergens, and other airborne particles. Additionally, consider systems with humidity control options to maintain optimal indoor humidity levels.

Example: An air conditioning system with a high-efficiency particulate air (HEPA) filter can significantly improve indoor air quality by capturing fine particles, such as pollen and dust mites, resulting in cleaner and healthier air for occupants.

i) Cost Considerations:

Evaluate the initial cost, installation cost, and long-term operating costs of the air conditioning system. Consider factors such as equipment cost, energy efficiency, maintenance requirements, and the availability of spare parts. A thorough cost analysis helps ensure that the selected system aligns with the budget and provides value for money.

Example: While a more energy-efficient air conditioning system may have a higher upfront cost, it can lead to significant long-term savings in energy bills, making it a cost-effective choice in the long run.

j) System Integration:

Consider the ability of the air conditioning system to integrate with other building automation systems, such as lighting or occupancy sensors. Integration allows for centralized control and optimization of energy usage, enhancing overall building efficiency and convenience.

Example: Integrating the air conditioning system with a building management system can enable features such as occupancy-based scheduling, where the system adjusts cooling settings based on room occupancy, leading to energy savings without compromising comfort.

By considering these factors, you can select a system that aligns with the specific requirements of the building while prioritizing comfort, energy efficiency, and cost effectiveness.

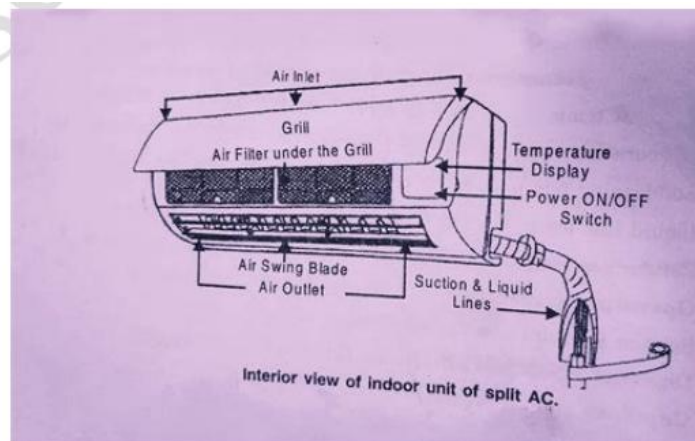
5. EQUIPMENT'S USED IN AIR CONDITIONING SYSTEM

a) Circulation Fan:

The circulation fan, also known as the blower or indoor fan, is responsible for moving air through the air conditioning system. It is typically located in the indoor unit and draws air from the room, passing it through the system for cooling or heating.

Function: The circulation fan helps circulate the conditioned air throughout the space, ensuring even distribution and maintaining comfortable temperatures.

▪ **Example:** In a split air conditioning system, the circulation fan is located in the indoor unit and blows air over the evaporator coil, where it is cooled before being distributed into the room through the supply ducts.

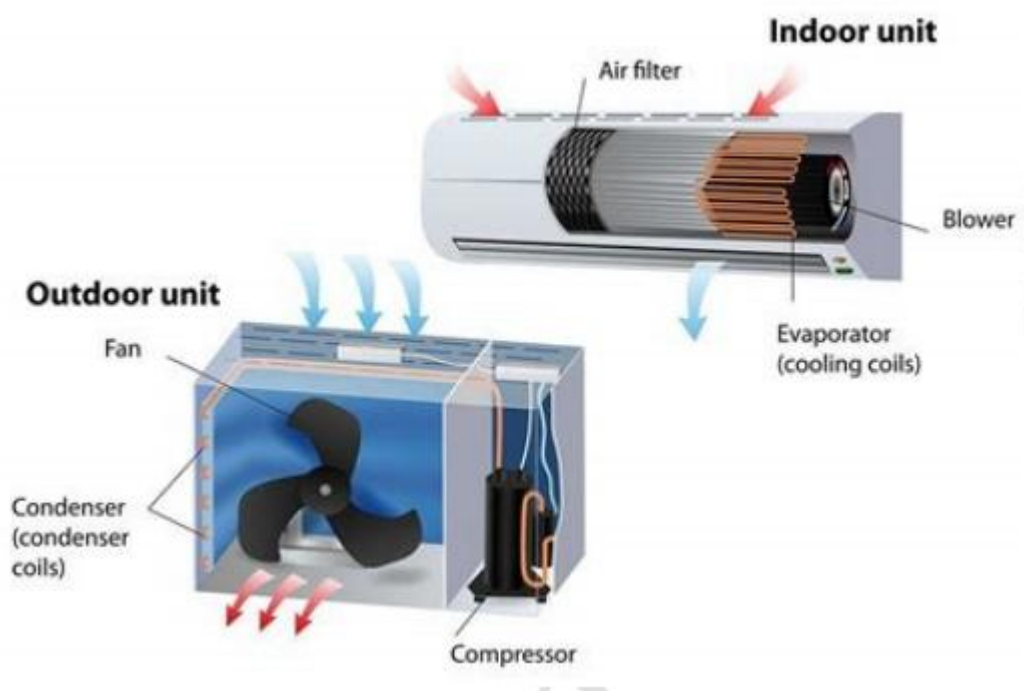


b) Air Conditioning Unit:

The air conditioning unit also referred to as the condensing unit or outdoor unit, is a vital component of most air conditioning systems. It is typically located outside the building or in a designated mechanical area.

Function: The air conditioning unit houses the compressor, condenser coil, and other components necessary for the heat exchange process. It facilitates the transfer of heat from the indoor space to the outdoor environment.

▪ **Example:** In a split air conditioning system, the air conditioning unit contains the compressor and condenser coil. It receives the refrigerant from the indoor unit, compresses it, and releases heat to the outside air.

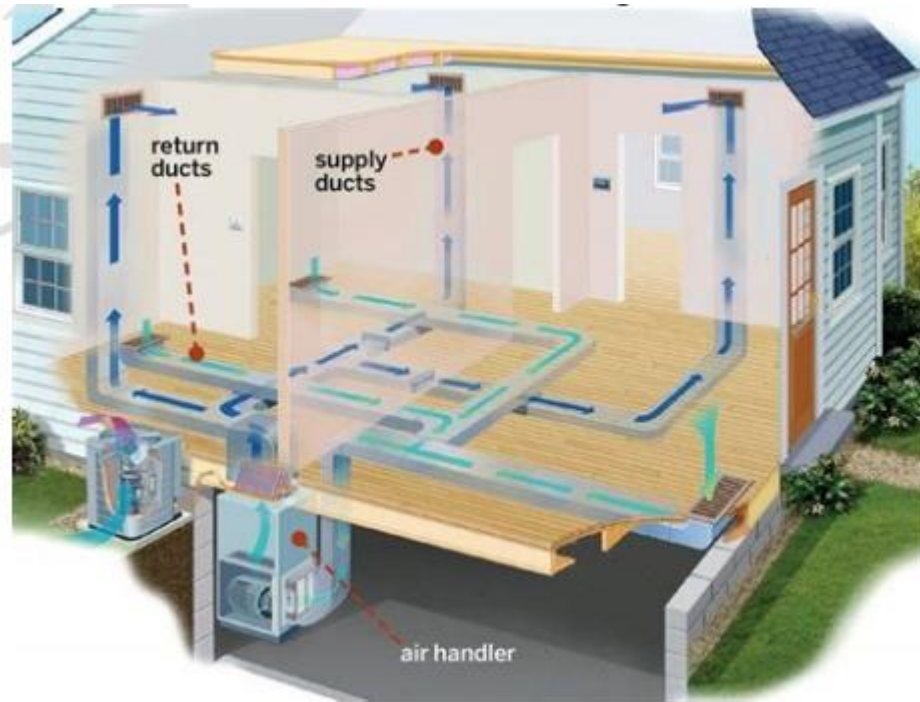


c) Supply Ducts:

Supply ducts are a network of channels or pipes that distribute conditioned air from the air conditioning unit to different areas or rooms within a building. These ducts are typically made of insulated metal or flexible material.

Function: Supply ducts transport the cooled or heated air from the air conditioning unit to the supply outlets, ensuring that the conditioned air reaches various zones or rooms.

▪ **Example:** In a central air conditioning system, supply ducts are responsible for carrying conditioned air from the central unit to different areas of the building, such as offices or bedrooms.

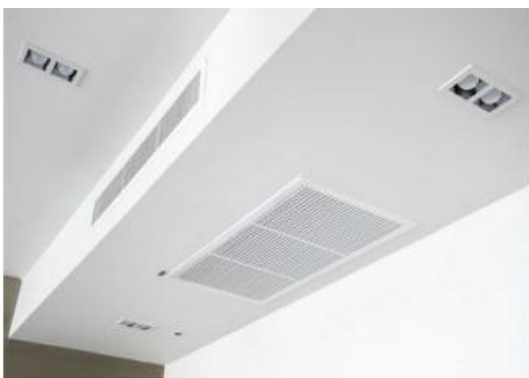


d) Supply Outlets:

Supply outlets, also known as registers or diffusers, are the points of air distribution within a space. They are typically located on walls, floors, or ceilings and allow conditioned air to flow into the room.

Function: Supply outlets disperse the conditioned air into the room, ensuring proper airflow and temperature distribution.

▪ **Example:** In a residential setting, supply outlets can be in the form of air vents on walls or ceiling-mounted diffusers. These outlets release the cooled or heated air into the room, providing comfort to the occupants.



e) Return Outlets:

Return outlets, also known as return vents or grilles, are openings that allow air to flow from the room back into the air conditioning system. They are typically located on walls or ceilings.

Function: Return outlets facilitate the return of air from the room to the air conditioning unit for reconditioning.

▪ **Example:** Return outlets are commonly found in residential or commercial spaces and ensure proper air circulation by allowing air to flow back to the system, where it undergoes the cooling or heating process again.

f) Filters:

Filters are an essential component of an air conditioning system as they help to improve indoor air quality by trapping dust, debris, allergens, and other airborne particles.

Function: Filters remove pollutants from the circulating air, resulting in cleaner and healthier indoor air.

▪ **Example:** Many air conditioning systems use filters, such as fibreglass filters or high-efficiency particulate air (HEPA) filters. These filters capture particles as air passes through them, preventing them from recirculating in the space.



These equipment components work together to ensure the proper functioning and effectiveness of an air conditioning system.

6. CLASSIFICATION OF AIR CONDITIONING SYSTEMS

According to purpose

- Comfort air conditioning.
- Industrial air conditioning

According to the season of the year

- Winter air conditioning.
- Summer air conditioning.
- Year-round (All weather air conditioning).

According to equipment arrangement

- Central air conditioning.
- Unitary air conditioning

According to the working substance used (Fluid);

- All air system.
- Chilled water system.
- Air water system

1.Functional classification:

From the functional point of view, air conditioning may be of two types:

a. Comfort air conditioning:

In this, the system aims at giving maximum human comfort to the occupants/users of conditioned space.

Function: Comfort air conditioning systems aim to control and maintain a desired temperature, humidity, and air quality level for the occupants' comfort.

Example: A common example of a comfort air conditioning system is a split air conditioner. It consists of an indoor unit that contains the evaporator coil, circulation fan, and air filters, and an outdoor unit that houses the compressor and condenser coil. The system removes heat and humidity from the indoor air and circulates cooled air back into the room

b) Industrial air conditioning:

In this, the conditioning creates, controls and maintains such an environment inside the conditioned space, that would suit best the needs of the industry.

Function: Industrial air conditioning systems are designed to create and maintain specific temperature and humidity conditions required for industrial processes. These processes can include manufacturing, data centres, cleanrooms, server rooms, or any other industrial application.

Example: An example of an industrial air conditioning system is a cleanroom air conditioning

system. Cleanrooms require strict control over temperature, humidity, and air cleanliness to ensure the integrity of products or processes. These systems often incorporate advanced filtration, precise temperature and humidity control, and positive or negative pressure control to meet the specific requirements of the cleanroom environment.

2.Classification based on season/temperature

Depending on the processes required for conditioning of outside air, air conditioning systems are classified in to:

a. Summer air conditioning:

In summer, the outside temperature is higher, and hence cooling of air is required for greater comfort. The cycle of operations consists of

- Air cleaning,
- Air cooling,
- Dehumidification, and
- Air distribution/circulation.

In this system, air is cooled & dehumidified. The outside air flows through the damper and is filtered using the filter. The air is then passed through a cooling coil and the air gets cooled below the required temperature and is also dehumidified. The cooled air is then passed through the water eliminator and water is collected in the sump. After that air is made to pass through a heating coil to reach the desired comfort condition.

The conditioned air is now supplied to the space using the blower.

b. Winter air conditioning:

In winter, the outside temperature is low, and hence heating of air is needed for comfort. The cycle of operations consists of

- Air cleaning,
- Air heating,
- Humidification and
- Air distribution/circulation

In this system, the outside air is sucked in through the damper and is filtered using an air filter. The air is then passed through a preheat coil to heat the air to control the evaporation of water in the humidifier. Steam is mixed with the air in the required quantity to increase the humidity to 55% RH. The air is then passed through a reheat coil to bring it to the comfort temperature of 22 °C. The conditioned air is then supplied to the conditioned space by a blower. Cold polluted air is discharged out and outside air is again sucked to repeat the cycle. These systems are used to heat the air and increase the humidity to acceptable levels.

c. Composite (All-Weather) Air Conditioning:

In this, the same air conditioning is done throughout the year, irrespective of outside temperature.

Function: Year-round air conditioning systems combine the capabilities of cooling and heating to maintain a comfortable indoor environment in any weather condition.

Example: Heat pumps are a common example of a year-round air conditioning system. They can provide both cooling and heating by reversing the refrigeration cycle. In summer, the heat pump operates like a regular air conditioner, removing heat from the indoor air. In winter, it

reverses the cycle, extracting heat from the outdoor air (or from the ground in geothermal heat pump systems) and transferring it inside to warm the indoor space.

3. According to Equipment Arrangement.

a) Central Air Conditioning:

Central air conditioning systems involve a centralized unit that provides cooling or heating to multiple areas or rooms through a network of ducts and vents

Function: Central air conditioning systems use a central unit, typically located outside the building, that cools or heats the air. The conditioned air is then distributed through ductwork and delivered to various zones or rooms using vents or diffusers.

Example: A common example of a central air conditioning system is a rooftop package unit. These units contain all components, including the compressor, condenser, and evaporator, in a single outdoor unit. They connect to a network of ducts that distribute the conditioned air to different areas or rooms within the building.

b) Unitary Air Conditioning:

Unitary air conditioning systems refer to self-contained units that provide cooling or heating to a single area or room.

Function: Unitary air conditioning systems are designed to provide localized cooling or heating for specific spaces. They are often installed directly in windows, walls, or as standalone units in the room.

Example: Window air conditioners are a common example of unitary air conditioning systems. These units consist of a single housing that contains all components, including the compressor, condenser, evaporator, and fan. They are typically installed in windows or openings in walls and provide cooling for a single room or area.

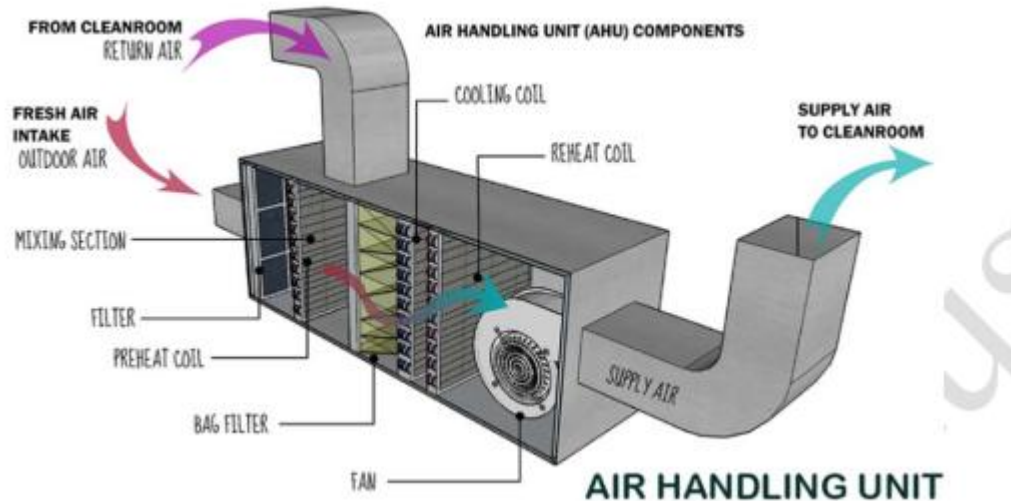
4. According to Working Substance Used (Fluid):

a) All Air System:

An all-air system utilizes air as the working substance for both cooling and heating.

Function: All-air systems use air as the medium to transport heating or cooling energy throughout the building. They typically incorporate air handlers, fans, and ductwork to distribute conditioned air to different zones or rooms.

Example: A common example of an all-air system is a forced-air furnace and air conditioner combination. The furnace heats the air by burning fuel or using electric resistance, and the air conditioner cools the air by removing heat. The conditioned air is then distributed through ducts to provide heating or cooling to the desired areas.



b) All Water System:

A chilled water system uses water as the working substance for cooling.

Function: Chilled water systems circulate chilled water through coils or pipes to remove heat from the indoor air. The chilled water absorbs heat from the surrounding air and is then cooled again in a chiller unit, typically located outside the building.

Example: A common example of a chilled water system is a central air conditioning system that utilizes chilled water coils. The chilled water flows through the coils, and the air is blown across them to remove heat, providing cooling to the space.

c) Air-Water System:

An air-water system combines the use of air and water as working substances.

Function: Air-water systems incorporate air handlers and water-based cooling or heating equipment to provide conditioned air. They can use water as a medium to transfer heat to or from the air in the system.

Example: An example of an air-water system is a water source heat pump system. It uses a water loop to transfer heat between the indoor air and the water source, such as a well, lake, or cooling tower. The system can provide both heating and cooling by extracting or releasing heat through the water loop.